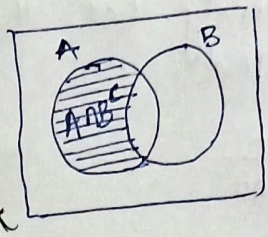
Quiz-1

(1). Since A and B are two independent events

$\Rightarrow P(A \cap B) = P(A)P(B)$. — (I) $\longrightarrow$ [redacted]



$\because (A \cap B^c) \cup (A \cap B) = A$. $\longrightarrow$ [redacted]

$\Rightarrow P(A \cap B^c) + P(A \cap B) = P(A)$. $\longrightarrow$ [redacted]

$$\therefore P(A \cap B^c) = P(A) - P(A \cap B)$$
$$= P(A) - P(A)P(B) \quad (\text{from (I)})$$
$$= P(A) \cdot (1 - P(B)) \quad \cancel{= P(A)(P(B)}$$

[redacted]

$$\boxed{P(A \cap B^c) = P(A)P(B^c)}$$

Hence A & B^c are independent events.

(2) Total no. of students in university, $n(\Omega) = 1200$.

Undergrad. students, $n(U) = 700$.

Grad. ~~stdn~~ students, $n(G) = 1200 - 700 = 500$.

Given, every student voted in campus referendum.

80 undergrad students voted in favor of the proposal.

$\Rightarrow (700 - 80 =)$ 620 undergrad students voted against the proposal.

and 60 graduate students voted against the proposal

$\Rightarrow (500 - 60 =)$ 440 grad students voted in favor of the proposal

[redacted]

Total no. of students who voted in favor $= 80 + 440 = 520$.

Let A = selected student voted in favor of the proposal

B = selected student is graduate

[redacted]

we have to find $P(B|A)$.

we know that

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$ [redacted]

$P(A)$ = Probability that the selected student ~~is~~ voted in favor of proposal

$$= \frac{520}{1200}.$$ [redacted]

$P(B \cap A)$ = Probability that the selected student is graduate & voted in favor of proposal

$$= \frac{440}{1200}$$ $\longrightarrow$ [redacted]

$$\therefore P(B|A) = \frac{440/1200}{520/1200} = \frac{44\cancel{0}}{52\cancel{0}} = \frac{11}{13}$$ [redacted].